

What is Computational Social Science?

SENSS Specialist Training: Introduction to Computational Methods

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Part 1

What is Computational Social Science?

- A discipline: a scientific field
- Interdisciplinary
- Key paradigms and methods

What it is not?

- Software engineering or software developer
- Simply a bag of tools
- Simply Social Science with powerful computers

Let's unpack each word in CSS

Computational Social Science

Scientific method: a disciplined and systematic way of thinking

- asking questions
- approaching questions and attempting to answer them
- evaluating (provisional) answers
- reformulating questions

The goal of science is to make the wonderful and complex understandable and simple—but not less wonderful.

—Herb Simon, *Sciences of the Artificial*

Computational Social Science

- What does social mean?
- What makes something social?

Computational Social Science

- Interactions and interdependence!
- No one is an isolated individual:
 - no entity is an isolated entity
- Context is shaped by past and present interactions and interdependencies

Computational Social Science

- Computation: transformation of information according to some rules
- a rule-based process that takes an input, performs operations on it, and produces an output
- Turing (1936): computation is any process that can be carried out by following a finite list of precise rules
- If you can specify the procedure, it is a computation. If you cannot, it is not!



Computation can be carried out on any medium:

- a silicon chip (electronic computers)
- biological cells and neurons
- piece of paper or a board game
- a room of people passing notes

Computation is the process not the medium

Computation is information processing

- Computation is information processing carried out by following precise rules
- Not all information processing is computation; but all computation is information processing
- Information processing paradigm:
 - Cells, individuals, organisations, and computers do it
 - Computation: a tool for studying societies (or cells, organizations, species)

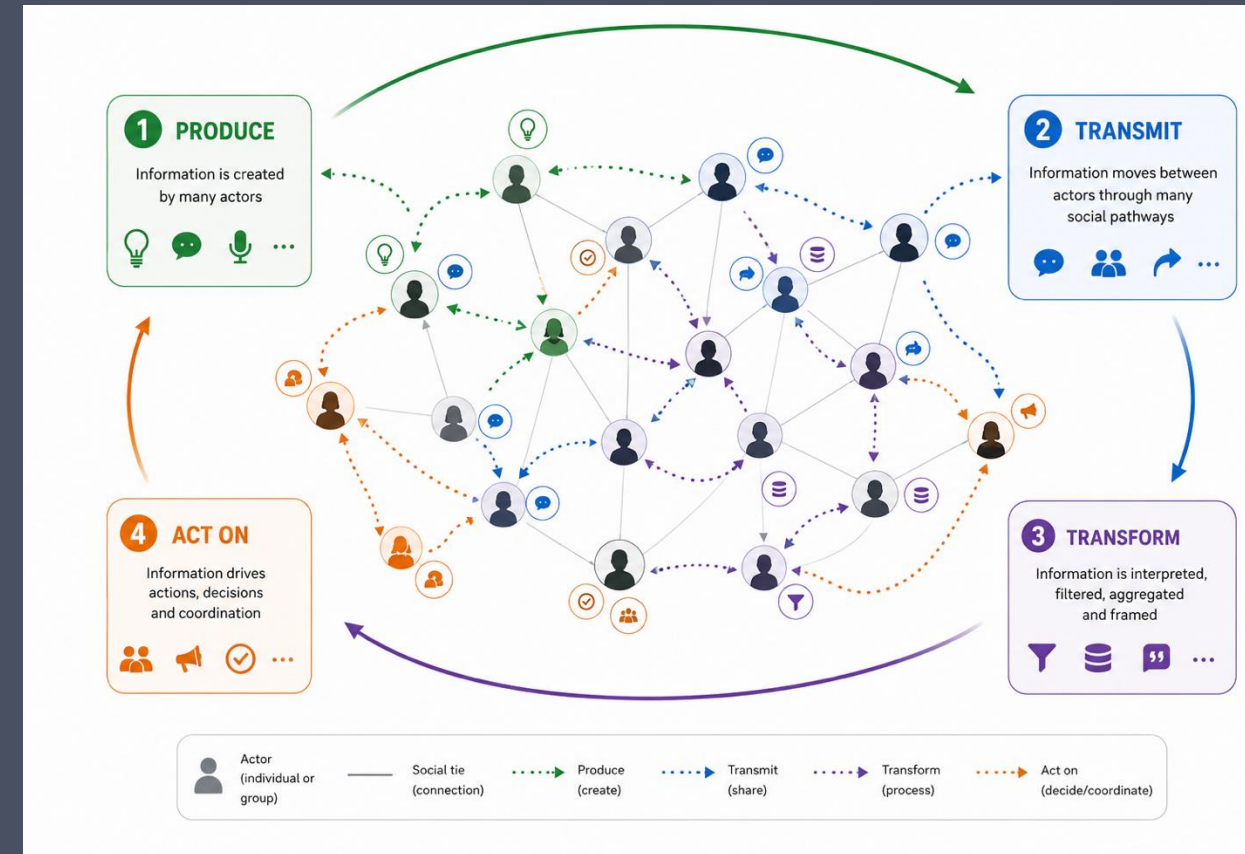
Information Processing Paradigm

- Societies are systems that produce, transmit, transform, and act on information
- Entities (i.e., agents) as information processors

Why this matters:

- **Ontologically**: this is how society actually operates
- **Epistemologically**: this is how we can study it

Sounds obvious?... but *information theory* influences CSS



Computational Social Science

Foundational ideas for CSS:

- Theory of computation (Alan Turing)
- Theory of communication (Shannon–Weaver model)

Information as signal:

- signal, noise, channel, transmission limit
- agents as information processors

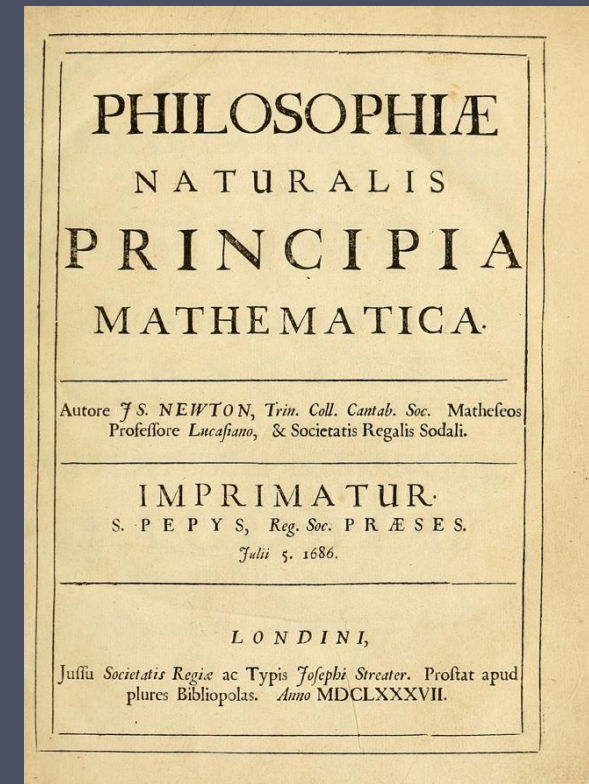
Application of these ideas to *social* settings → CSS

- *Herbert A. Simon. The Sciences of the Artificial, 1969*

A wealth of information creates a poverty of attention (Herb A. Simon)

An Instrument-Enabled Field

- Science is always enabled and revolutionized by instruments
- We create and use tools to create new science



CSS as An Instrument Enabled Field

- Computers are to social science what the telescope is to astronomy.

Digital revolution:

- Modern computers are extremely powerful
- So much digital data available

Data and Big Data

- We have moved from data-poor to data-rich social science
- Not only for new data: historical archives are being digitalized
- Digital trace data
- Tabular data vs non-tabular data;
 - real-time; repurposed data



Significant development: Phase Transition!

Data Poor → Data Rich

- Yes, we always had data, but data availability has entered a new era
- **Qualitative change:** *more of the same becomes something different*
- *A phase transition!*
- **Example:** photo and movie difference
- Karl Marx citing Hegel: "a certain point merely quantitative differences pass over by a dialectical inversion into qualitative distinctions."

Key Concepts in CSS

Key Concepts in CSS

- Information Processing Paradigm
- Complexity

Social Sciences: deal with complex adaptive systems

Complexity in a nutshell

*A system is **complex** when its behaviour **emerges** from the interactions of its parts rather than from the parts themselves.*

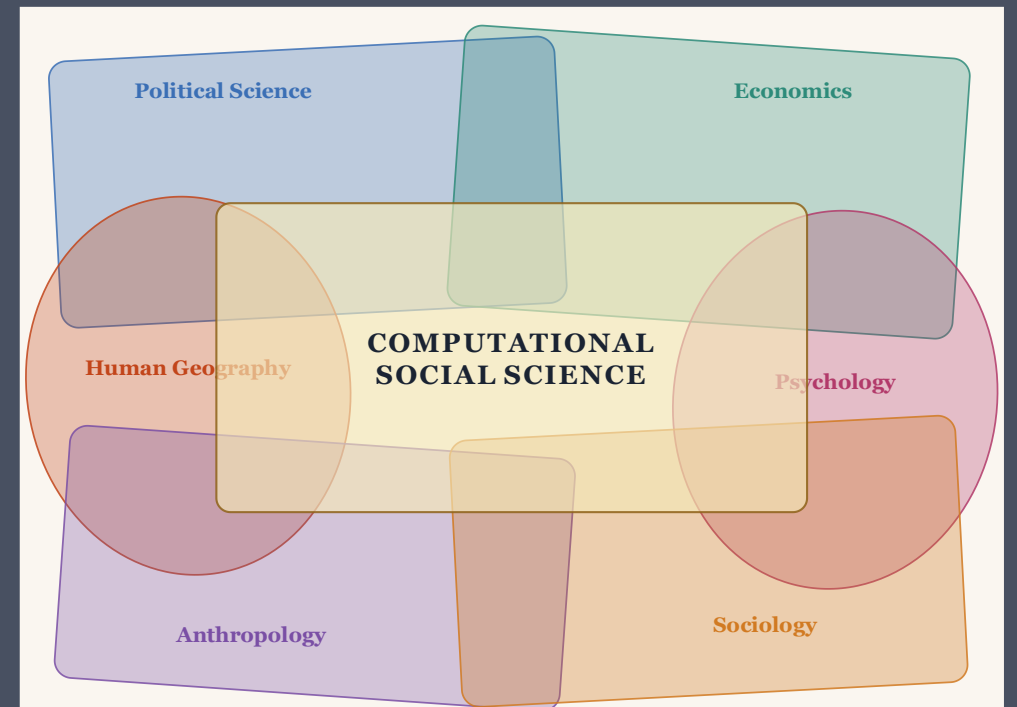
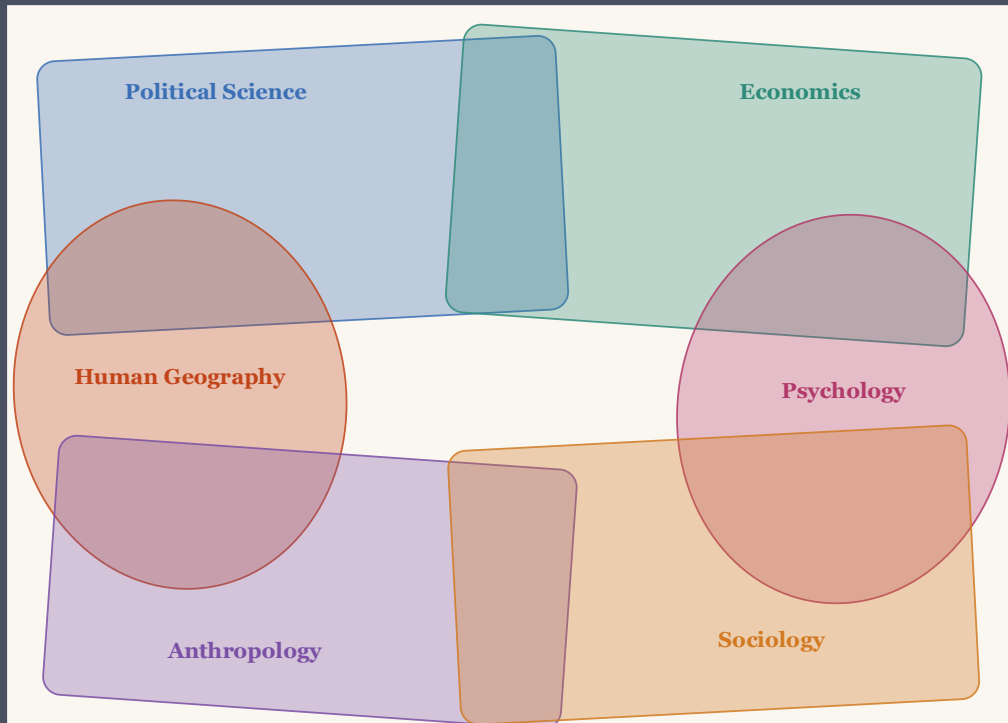
- *emergence*

Examples:

- Adam Smith's invisible hand
- Traffic-jam
- Starling murmuration

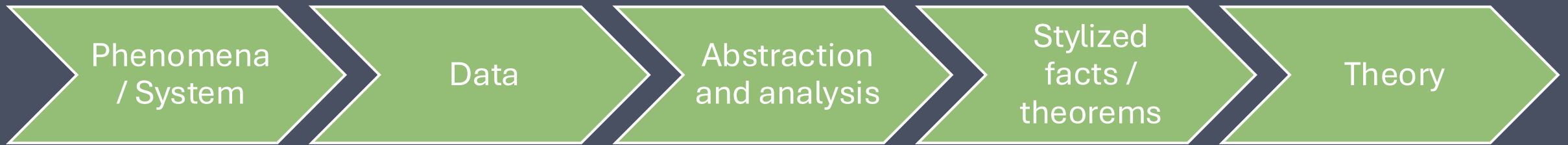
Mapping CSS

Computational Social Science as a Field

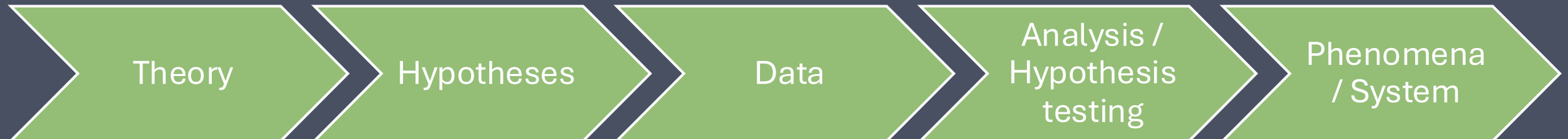


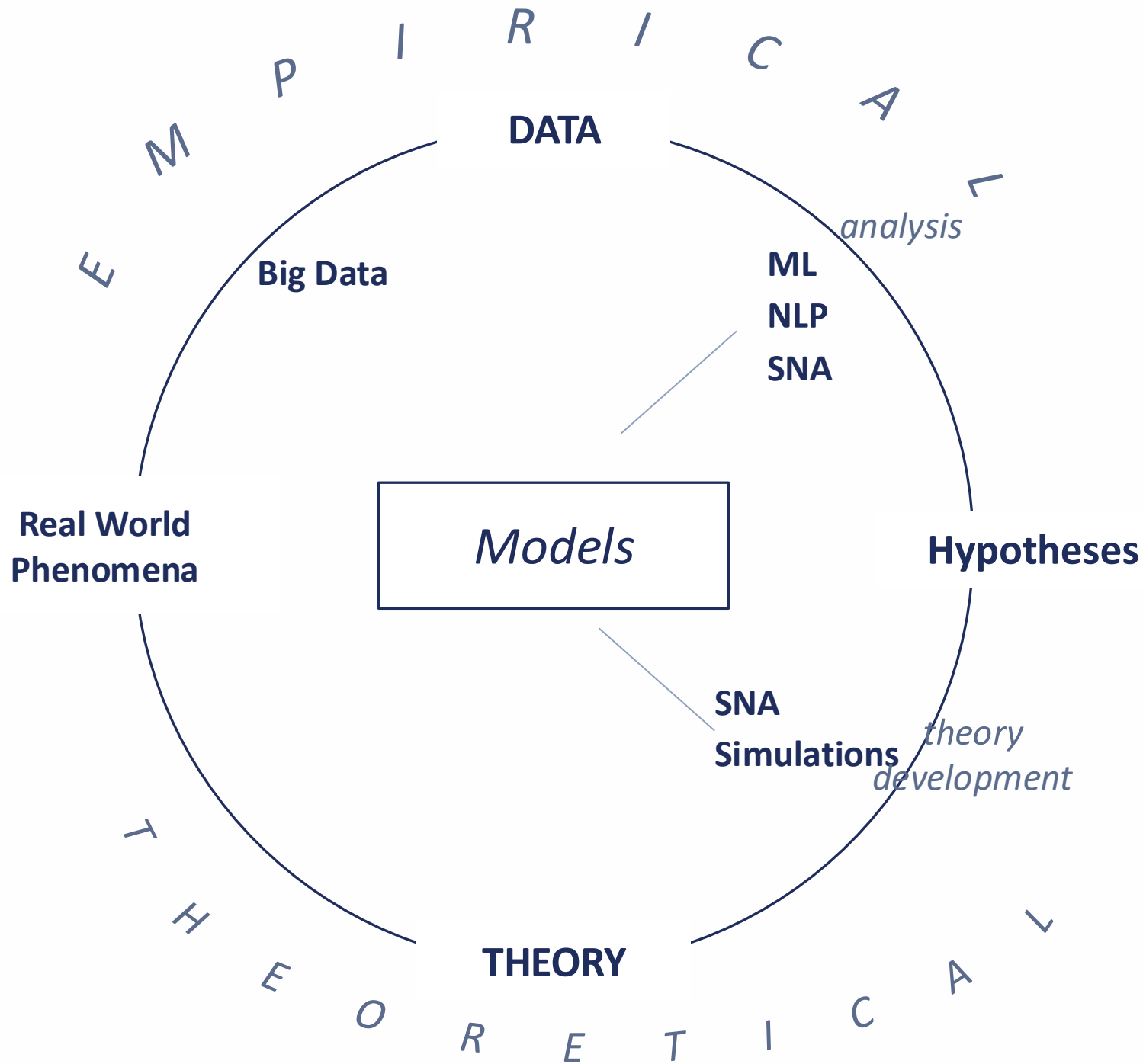
Scientific Method: Simplified Sketch

Induction:



Deduction:





Two domains of scientific inquiry

THEORETICAL

Building artificial universes

- Agent-based simulations
- Social Network Analysis

EMPIRICAL

Processing and analysing data

- Automated data collection
- Machine Learning
- Natural Language Processing
- Social Network Analysis

Which field is easier to deal?
Physics or Sociology?